

G Lead 8

An eight-voice analogue-style polysynth for leads and poly parts, with a built-in arpeggiator.



Overview

G Lead 8 is a classic subtractive synth: two oscillators per voice into a warm 24 dB ladder filter, with its own envelopes for the filter and the amp, and an LFO for vibrato, pulse-width and filter movement. It plays up to eight notes at once, or one at a time in MONO for legato leads with glide.

A built-in arpeggiator turns held chords into tempo-locked patterns, and twelve patches give you starting points from classic leads to brass, strings and wobble.

GFX drift keeps it alive: the tuning wanders slowly, each voice's oscillators drift apart a little on their own, each voice's pulse width wanders, and every note lands slightly differently in tune. The voices are spread a little across the stereo field.

Reach for it for

- Classic mono synth leads with glide
- Poly brass, strings and soft pads
- Arpeggiated parts locked to the project
- PWM and detuned textures
- Resonant filter sweeps and wobbles

Playing it

1. Insert G Lead 8 on a track as an instrument (it has no audio input; it plays from MIDI).
2. Arm the track for recording with a MIDI input and turn on record monitoring, or draw notes in a MIDI item.
3. The MIDI light in the header blinks green when notes arrive. If you use more than one MIDI channel, set CHANNEL to the one you want; OMNI plays on all.
4. Pitch bend covers +/-2 semitones, the mod wheel (CC1) adds vibrato, the sustain pedal (CC64) holds notes, and CC123 / CC120 stop all notes.

Controls

The header

Patch display	Shows the patch number and name, POLY or MONO, how many of the eight voices are sounding, and ARP ON when the arpeggiator is running. Click it for the list of patches.
< / >	Step to the previous or next patch.
MIDI	Blinks green when a note arrives.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

OSCILLATORS

SAW / PULSE (OSC 1, OSC 2)	The waveform of each oscillator. SAW is bright and buzzy; PULSE is hollower and changes with PW.
COARSE	Oscillator 2's pitch in semitones, -24 to +24. +7 gives fifths, +12 or -12 an octave.

FINE	Oscillator 2's fine tuning, -50 to +50 cents. A few cents of detune against oscillator 1 gives the thick, beating analogue sound.
MIX	The balance between the two oscillators, from OSC 1 only to OSC 2 only.
PW	Pulse width for the PULSE waves, 5 to 95%. 50% is a square; narrower settings are thinner and more nasal.

FILTER

CUTOFF	The ladder filter's cutoff frequency, 20 Hz to 18 kHz. Lower is darker.
RESONANCE	0 to 100%. Emphasises the frequencies around the cutoff, from smooth to squelchy; near the top it starts to ring.
ENV	How far the filter envelope moves the cutoff, -100% to +100% (up to six octaves). Negative values close the filter instead of opening it.
KEY	Key tracking, 0 to 100%. At 100% the cutoff follows the keyboard, so high notes are as bright as low ones.

FILTER ENVELOPE and AMP ENVELOPE

Both envelopes have the same four controls. The filter envelope moves the cutoff (by the ENV amount); the amp envelope shapes the volume of each note.

ATTACK	How fast it rises when a key goes down, 0.5 ms to 3 s.
DECAY	How fast it falls to the sustain level, 5 ms to 5 s.
SUSTAIN	The level held while the key is down, 0 to 100%.
RELEASE	How fast it dies away after the key is released, 5 ms to 5 s.

LFO

SHAPE	SINE, TRI, SAW, SQUARE or S&H (sample and hold: a new random value each cycle).
RATE	The LFO speed, 0.05 to 20 Hz.
VIBRATO	How much the LFO moves the pitch, 0 to 100 cents. The mod wheel adds up to another 50 cents on top.
PWM	How much the LFO moves the pulse width, 0 to 100%. Only heard on PULSE waves.
CUTOFF	How much the LFO moves the filter cutoff, 0 to 100% (up to three octaves each way) - for wah and wobble.

ARPEGGIATOR

The GFX arpeggiator, built in. With ARP on, held notes play as an arpeggio, every step on. It locks to the project tempo when the project plays and runs free at the project tempo from the first key when it is stopped.

ARP	Switches the arpeggiator on (red light) or off. Switching it stops any notes it was playing.
LATCH	On: the arpeggio keeps playing after you let go. Release all keys and play a new chord to replace it.
PANIC	Stops every note at once.
MODE	UP, DOWN, UP-DN, DN-UP, PLAYED (the order you played the notes), RANDOM or CHORD (the whole chord on every step).
RATE	The step length: 1/4, 1/8, 1/8T, 1/16, 1/16T or 1/32 (T = triplet).
OCTAVES	1 to 4: how many octaves upwards the arpeggio repeats the notes.
GATE	How long each note is held, 10 to 200% of its step. Above 100% the notes overlap.

SWING

0 to 75%: lengthens the first step of each pair and shortens the second.

PERFORMANCE

POLY / MONO

POLY plays up to eight notes. MONO plays one note at a time, legato: a new note held over an old one glides to it without a new attack, the last note wins, and letting go returns to a note still held. Switching stops all notes.

GLIDE

Portamento time, 0 to 1000 ms. In MONO the pitch slides from note to note; in POLY each voice slides from the last note it played.

VELOCITY

0 to 100%: how much playing harder makes notes louder and opens the filter envelope further. 0 ignores velocity.

DRIFT

0 to 100% (25% by default). At 100% the whole instrument's tuning wanders by up to 20 cents, each voice's oscillators drift apart by up to 8 cents and its pulse width wanders, and each note is struck up to 6 cents out of tune. 0 turns it all off.

VOLUME

Output level, -24 to +6 dB.

CHANNEL

The MIDI channel it listens to: OMNI (all) or 1 to 16.

Patches

Choosing a patch sets the oscillators, filter, both envelopes, the LFO, GLIDE and POLY / MONO; the arpeggiator, VELOCITY, DRIFT, VOLUME and CHANNEL are left as they are. Every patch is level-matched so a single-note line plays at about the same loudness.

Classic Lead

Two saws a few cents apart, a moderate filter sweep and a little vibrato; mono.

Poly Brass

Detuned saws with a slow-opening filter and soft attack: synth brass; poly.

Fat Mono	Saws an octave apart with more resonance and a 60 ms glide; mono.
Square Lead	Two square waves an octave apart, bright, with sine vibrato and glide; mono.
PWM Strings	Detuned pulses with a slow LFO on the pulse width, slow attack and long release; poly.
Hollow Pipe	Narrow pulses an octave apart with resonance and vibrato: a reedy pipe; mono.
Fifths	Two saws a fifth apart with glide; mono.
Soft Poly	Saw and pulse through a darker filter with a falling amp envelope; poly.
Resonant Sweep	A low, highly resonant filter swept hard by its envelope; mono.
Vintage Detune	Saws 18 cents apart with gentle vibrato and a longer release; poly.
Wobble Lead	Saws an octave apart with a 4 Hz LFO on the cutoff; mono.
Init	A plain starting point: two slightly detuned saws; poly.

Tips

- For a classic lead, use MONO with a little GLIDE and leave VIBRATO at zero - then add vibrato with the mod wheel as you play.
- For slowly opening swells, raise the filter envelope's ATTACK and set ENV well above zero with CUTOFF low.
- The output stage (SAT and OFF / CLIP / LIMIT, CLIP by default) and the stereo controls (mono below a frequency, width) have no knobs on the panel; you can reach them in REAPER's parameter list.
- With the arpeggiator on, try CHORD mode with a short GATE for rhythmic stabs.

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